System Analysis and Design

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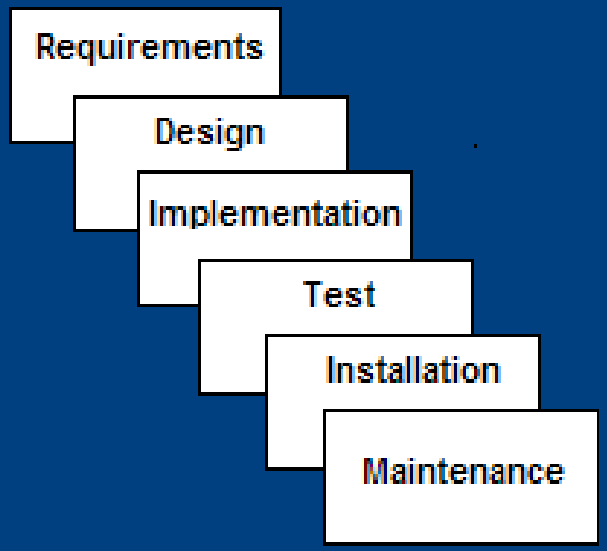
Waterfall Model



- **Requirements** – defines needed information, function, behavior, performance and interfaces.
- **Design** – data structures, software architecture, interface representations, algorithmic details.
- **Implementation** – source code, database, user documentation, testing.

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Waterfall Strengths

- Easy to understand, easy to use
- Provides structure to inexperienced staff
- Milestones are well understood
- Sets requirements stability
- Good for management control (plan, staff, track)
- Works well when quality is more important than cost or schedule

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Waterfall Deficiencies

- All requirements must be known upfront
- Deliverables created for each phase are considered frozen – inhibits flexibility
- Can give a false impression of progress
- Does not reflect problem-solving nature of software development – iterations of phases
- Integration is one big bang at the end
- Little opportunity for customer to preview the system (until it may be too late)

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When to use the Waterfall Model

- Requirements are very well known
- Product definition is stable
- Technology is understood
- New version of an existing product
- Porting an existing product to a new platform.
  - High risk for new systems because of specification and design problems.
  - Low risk for well-understood developments using familiar technology.

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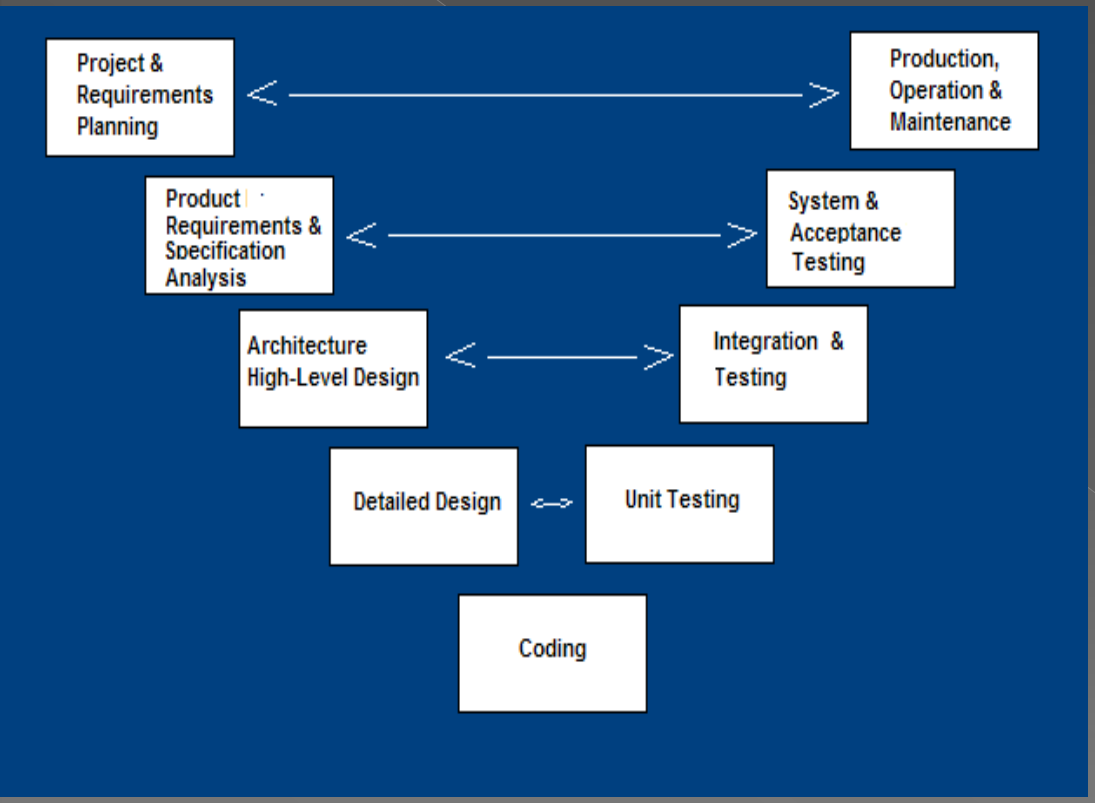
V-Shaped SDLC Model



- A variant of the Waterfall that emphasizes the verification and validation of the product.
- Testing of the product is planned in parallel with a corresponding phase of development

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V-Shaped Steps

- Project and Requirements Planning – allocate resources
- Product Requirements and Specification Analysis – complete specification of the software system
- Architecture or High-Level Design – defines how software functions fulfill the design
- Detailed Design – develop algorithms for each architectural component
- Production, operation and maintenance – provide for enhancement and corrections
- System and acceptance testing – check the entire software system in its environment
- Integration and Testing – check that modules interconnect correctly
- Unit testing – check that each module acts as expected
- Coding – transform algorithms into software

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V-Shaped Strengths

- Emphasize planning for verification and validation of the product in early stages of product development
- Each deliverable must be testable
- Project management can track progress by milestones
- Easy to use

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V-Shaped Weaknesses

- Does not easily handle concurrent events
- Does not handle iterations or phases
- Does not easily handle dynamic changes in requirements
- Does not contain risk analysis activities

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When to use the V-Shaped Model

- Excellent choice for systems requiring high reliability – hospital patient control applications
- All requirements are known up-front
- When it can be modified to handle changing requirements beyond analysis phase
- Solution and technology are known

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Agile software development life cycles

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Agile SDLC's

- Speed up or bypass one or more life cycle phases
- Usually less formal and reduced scope
- Used for time-critical applications
- Used in organizations that employ disciplined methods

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Some Agile Methods

- Rapid Application Development (RAD)
- Incremental SDLC
- Scrum
- Extreme Programming (XP)
- Adaptive Software Development (ASD)
- Feature Driven Development (FDD)
- Crystal Clear
- Dynamic Software Development Method (DSDM)
- Rational Unify Process (RUP)

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Agile vs Waterfall Propaganda

- http://www.youtube.com/watch?v=gDDO3ob-4ZY&feature=related

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WHAT IS A PROJECT?

- "*A project is a problem scheduled for solution.*" *This definition forces us to recognize that projects are aimed at solving problems and that failure to define the problem properly is what sometimes gets us into trouble.*

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What is the problem?

- **A desired objective is not a problem by itself.** ***The key to a problem is that there is an obstacle that prevents you from closing the gap.***
- **A problem is a** ***gap(achieving your objective)*** **between where you are and where you want to be, with an obstacle that prevents easy movement to close the gap.**
- **Problem solving consists of finding ways of overcoming or getting around obstacles.**

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WHAT IS PROJECT MANAGEMENT?

- **Project management is the planning, scheduling, and controlling of project activities to meet project objectives.**
- **The major objectives that must be met include performance, cost, and time goals, while at the same time you control or maintain the scope of the project at the correct level.**

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The scope of project

- The scope of a project should remain constant throughout the life of the job.
- Unforeseen problems or an inadequately defined problem the most common reason for scope changes is that something is forgotten.
- In most cases the magnitude (scope) of the work increases, as a result of overlooked details.

Continue to see more…

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...*The scope of project*

- **Scope generally increases.**
  The only time project scope decreases is when the budget is cut, and some of the originally planned work is put on hold.
- **The problem with scope changes is that they tend to be small and incremental, if a number of them occur, the project budget or schedule may suffer. This is a fairly common cause of project failures.**

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Project Manager & The Scope

- A project manager has a responsibility to keep stakeholders informed about the impact of scope changes on the project, protecting them from surprises at the end of the job and protecting the project manager from being evaluated on original targets rather than on revised ones.

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The Four Project Objectives are…

- *Performance*
- *scope*
- *Cost*
- *Time*

Continue…There are more!

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Performance, scope, Cost& Time

- ***Performance:*** **The quality of the work being done.**
- ***Scope:*** **The magnitude of the work to be performed.**
- ***Cost:*** **The cost of project work, directly related to the human and physical resources applied.**
- ***Time:*** **The schedule that must be met.**

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The relation between the four project objectives:

$$Cost=f(P,T,S)$$

To understand this eq. go on…

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Continue…What did the equation say?

- cost is a function (*f*) of performance (*P*), time (*T*), and scope (*S*). As *P* and *S* increase, cost generally increases.
- The relationship between time and cost, however, is not linear. As a rule, cost increases as the time to do the project decreases below a certain optimum time.
- If the duration is shortened, it is often necessary to pay premium labor rates as a consequence. Further, worker errors often increase, resulting in costs for corrections, and productivity often declines. Move…

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Comment:

- **Enough people are thrown at a project, it can be completed in whatever time is desired. This is simply not true. but the idea is the cause of many project fiascos.**

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THE HUMAN SIDE OF PROJECT MANAGEMENT

- **Factors***(components)***affect the success of a project:**
- ***The Right People***
- ***The Right Type of Management***

- ***One of the key ingredients is having the right people on the job and managing them appropriately.***

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STEPS IN MANAGING A PROJECT

- ***Define the problem***
- ***Develop solution options***
- ***Plan the project***
- ***Execute the plan***
- ***Monitor and control progress***
- ***Close the project.***

To see more about them the…↓

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The Project Management System

- In order to manage projects successfully, it is necessary to have a system. A full project management system consists of seven components.
- If any one of the seven components is not in place or does not function satisfactorily, then you will have some difficulty managing projects.

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The seven components are…

- ***Human Factors.***
- ***Method.***
- ***Culture.***
- ***Organization.***
- ***Planning.***
- ***Information.***
- ***Control.***

Continue…

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Human Factors

A project manager must be able to deal effectively with all of the parts of this subsystem in order to be successful.

- Leadership.
- Negotiation.
- Team building.
- Motivation.
- Communication.
- Decision making.

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Continue..The seven components

- **Methods** refer to the tools of your trade.
- **The culture** of an organization affects everything you do.
- ***Organization:*** Every organization must deal with the assignment and definition of each person's authority, responsibility, and accountability.
- ***Planning:*** Every organization needs a good methodology for planning projects if it is to be successful.

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Continue… Information & Control

- Good historical data are needed for planning projects.
- The control subsystem is supported by the planning and information subsystems.

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